

Junior Makers · Class 22

# 8×8 LED Matrix Emoji Display

64 lights, endless pictures

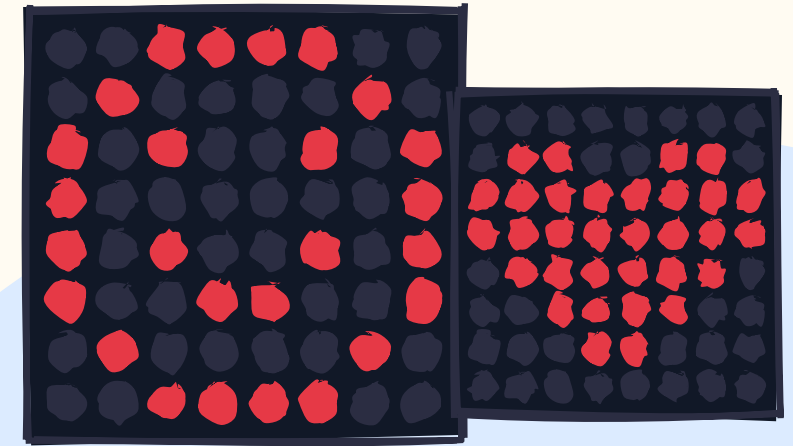
Rows & columns

Binary pictures

Animation

Did you know?

The scoreboard at a cricket stadium is a giant grid of LEDs, controlled the same way as our little 8×8 matrix.





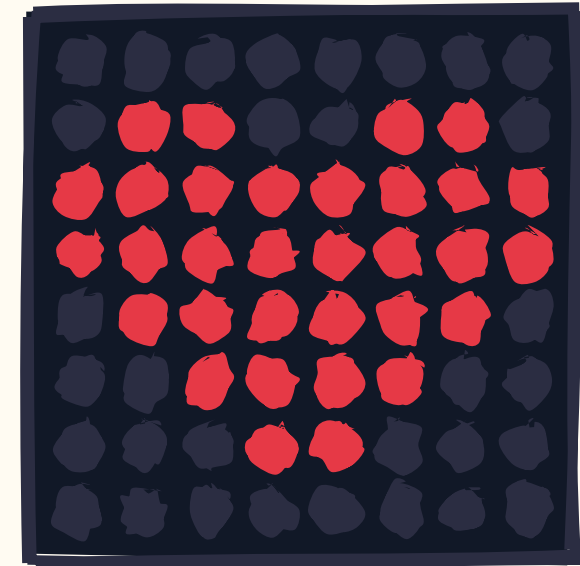
Let's think...

# How do bus signs show words?



## Dots everywhere

Bus signs, shop signs and stadium scoreboards show words, numbers and even pictures. Look closely: it's all **dots** that are either on or off. How do you tell 64 dots what picture to make?



One wire each?

Draw with a pen?

Rows of 0s and 1s!

**Today:** you design **emojis** and **animations** for an 8x8 LED matrix and scroll your own message.





## Learning objectives

# By the end of this class I can...



1

**Rows & columns:** find any LED on an 8×8 grid by its **row** and **column****Proof:** I can light row 2, column 5 when asked

2

**Pictures as bits:** turn a drawing into **8 rows of 0s and 1s****Proof:** my emoji drawing has 8 binary rows

3

**Animation:** make an **animation** by showing frames one after another**Proof:** my emoji blinks or moves

4

**Message board:** **scroll** a message and design a useful sign**Proof:** my matrix scrolls a message for our class



Objective 1 • Rows & columns • video

# Video time!



## How Pixels Work

In One Lesson • about 3 minutes • [youtu.be/0B79dGR19Tg](https://youtu.be/0B79dGR19Tg)

### Watch for...

- 1 What is a **pixel**?
- 2 Why do pictures look smooth when there are lots of pixels?
- 3 How does a screen make so many colours?



**After the video:** How many pixels does our matrix have? Would a photo of your face be recognisable on it?



## Objective 1 • Rows &amp; columns

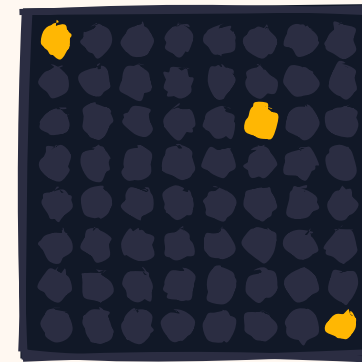
# Rows and columns, starting at 0

## The idea

The matrix has **8 rows** (across) and **8 columns** (down): 64 LEDs. Each LED has an address: **(row, column)**, numbered **0 to 7**.

Row 0 is the top. Column 0 is the left. So (0, 0) is the top-left corner and (7, 7) is the bottom-right.

## Example: seats in a cinema



Your cinema ticket says 'Row F, Seat 12'. With just two pieces of information you find your exact seat among hundreds. Every LED has a 'ticket' too: its row and column.



Class game

# Human matrix



## How to play

1. Make a  $4 \times 4$  grid of 16 students sitting on chairs.
2. Each holds a card: **bright side = ON**, dark side = OFF.
3. The 'computer' reads out rows as 0s and 1s: **Row 0: 0110!**
4. Students in that row show ON or OFF to match.
5. After 4 rows, what picture did you make?

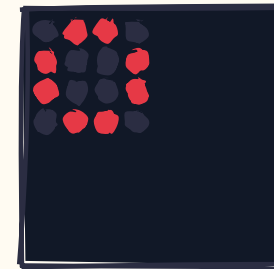
## Then discuss

How many numbers did the computer need to say to make a whole picture?



## Big idea

A picture can be sent as rows of 0s and 1s. That's how matrices, screens and printers work.



## Objective 2 · Pictures as bits

# A picture is 8 rows of bits

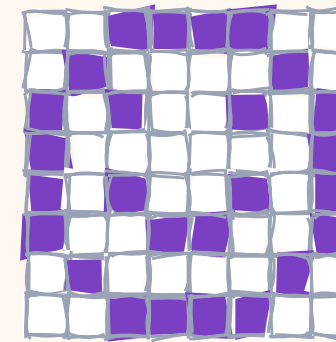
## The idea

Each row is 8 LEDs: **1 = on**, **0 = off**. So one row is 8 bits: **one byte** (remember JM-19's bracelets?).

A whole smiley is just **8 bytes**:

00111100 · 01000010 · 10100101 · 10000001 · 10100101  
· 10011001 · 01000010 · 00111100

## Example: cross-stitch patterns



00111100  
01000010  
10100101  
10000001  
10100101  
10011001  
01000010  
00111100

Embroidery patterns are grids: each square is either stitched or not. Your grandmother could copy a design just from a grid of filled and empty squares. That's exactly our emoji grid.

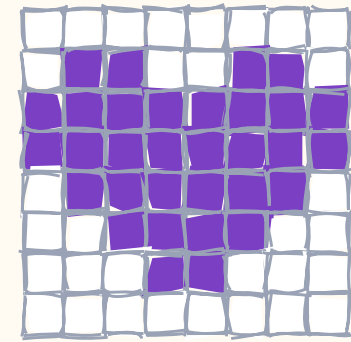


## Objective 2 · Build step 1 of 4

# Design your emoji on paper



- Take the 8×8 template. Shade squares to draw an emoji or icon.
- Keep it bold: thin details disappear on 64 dots.
- Write each row as 8 digits: shaded = 1, empty = 0.
- Check: every row must have exactly **8** digits.
- Swap with a partner: can they draw your emoji from just the numbers?



00000000  
01100110  
11111111  
11111111  
01111110  
00111100  
00011000  
00000000



**Check:** Your partner redraws your emoji perfectly from the 8 rows of numbers.



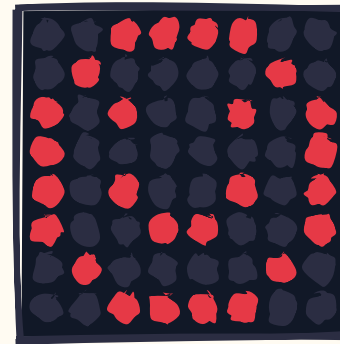


Objective 1 • Rows & columns • Build step 2 of 4

# Wire the matrix



| Matrix pin | Arduino |
|------------|---------|
| VCC        | 5V      |
| GND        | GND     |
| DIN        | pin 12  |
| CS         | pin 10  |
| CLK        | pin 11  |



## The MAX7219 chip

Controlling 64 LEDs would need 64 wires! The chip on the back does the work: we send it just 3 wires of data.





Objective 2 · Pictures as bits · Build step 3 of 4

# Show your emoji

## Setup

LED matrix init DIN 12 CLK 11 CS 10

LED matrix set intensity 4

## Loop

LED matrix draw image [8x8 grid editor: your emoji]

wait 2 seconds

LED matrix clear

wait 0.5 seconds

## Grid editor

In ArduinoBlocks, click the squares in the image block to copy your paper design. Or type your 8 binary rows.



## Upside down?

Modules are mounted differently. If your emoji is rotated, turn the matrix or flip the design.



## Objective 3 · Animation

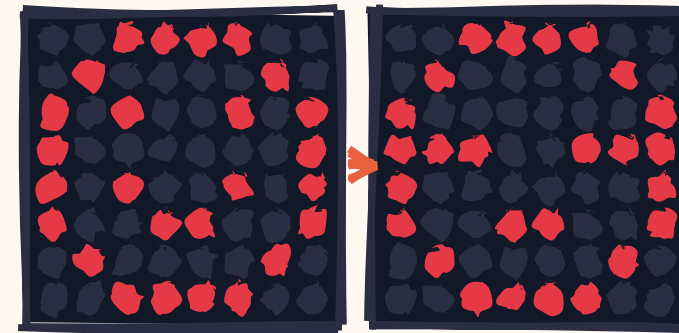
# Animation = frames shown fast

## The idea

A **frame** is one picture. Show frame 1, wait, show frame 2, wait... Your brain joins them into **movement**.

A **blink** needs just 2 frames: eyes open, eyes closed. A **bouncing ball** might need 6. Short waits (100–200 ms) look smooth; long waits look jumpy.

## Example: a flipbook



Draw a stick figure in the corner of every page of a notebook, each a tiny bit different. Flick the pages: it moves! Cartoons and games work the same way, with 24 to 60 frames every second.



Objective 3 · Animation · Build step 4 of 4

# Make it blink

## Loop

LED matrix draw image [face, eyes OPEN]

wait 2 seconds

LED matrix draw image [face, eyes CLOSED]

wait 150 milliseconds

(bonus) repeat 8 times: scroll image left by 1



## Timing

## Looks like...

Long wait open, short closed

Equal waits

Very short (20 ms)



## Real blinks

People blink for about 0.1 to 0.4 seconds. That's why 150 ms looks natural!



## Objective 4 · Message board

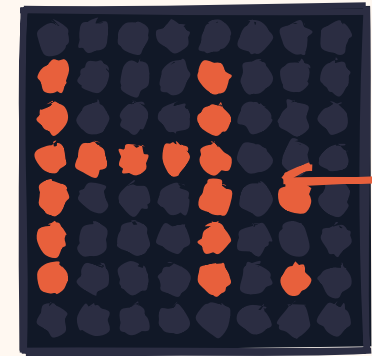
# Scrolling messages

## The idea

Words are too wide for 8 columns, so we **scroll**: shift every column one step left, add a new column on the right, repeat. ArduinoBlocks has a **scroll text** block that does this for you.

Choose the **speed** carefully: too fast and nobody can read it, too slow and people walk past.

## Example: a news ticker



TV news channels show a moving line of headlines at the bottom of the screen. You can read a long sentence in a small space because it slides past you.



Real world

# LED matrices all around



## Scoreboards

Stadium screens are huge  
LED grids



## Bus and train signs

Show the next stop and  
arrival time



## Shop signs

'OPEN' signs and moving  
price displays



## Clocks

Digital clocks and timers in  
airports



Design a sign for our school: what message would help people most?

## Pixels

Your phone screen is the same idea, with  
millions of tiny RGB dots instead of 64 red  
ones.



## Objective 4 • Message board • design

# Make a useful sign



| Sign         | Shows                        | Animation     |
|--------------|------------------------------|---------------|
| Welcome sign | scroll 'WELCOME TO ROBOTICS' | + smiley      |
| Quiet please | a 'shh' face                 | blinks slowly |
| Timer        | countdown 5, 4, 3, 2, 1      | then a star   |
| Your idea    |                              |               |



## Buzzer bonus

Beep the buzzer when the message starts, so people look!





## Test like an engineer

# Bug hunt



| Bug                     | Likely cause                             | Did it happen? |
|-------------------------|------------------------------------------|----------------|
| Nothing lights          | VCC/GND swapped or DIN/CLK/CS wrong pins |                |
| All 64 LEDs on          | matrix not initialised / wrong CS pin    |                |
| Emoji upside down       | module rotated                           |                |
| Very dim or very bright | intensity setting (0–15)                 |                |
| Scroll unreadable       | speed too fast                           |                |

### Debug checklist

Light just ONE LED at (0, 0) first. Then you know where the top-left corner really is.



Change one block at a time and test again.



Level up!

# Matrix challenges

## Choose your challenge



**Heartbeat:** a heart that grows and shrinks (3 frames).



**Pac-Man:** an animated mouth that opens and closes as it moves.



**Countdown:** show 9 down to 0, then a smiley and a beep.



**Rain:** dots fall from the top row to the bottom, at random columns.

## Think like an engineer

How many bytes do you need to store a 10-frame animation on an 8×8 matrix?

## Boss level

Join two teams' matrices side by side and scroll one long message across both!



## Quick quiz

# Bits and pictures



| Question                                         | Answer |
|--------------------------------------------------|--------|
| 1. How many LEDs on an 8×8 matrix?               |        |
| 2. One row of 8 LEDs is how many bits?           |        |
| 3. Write a row with only the two middle LEDs on. |        |
| 4. What address is the bottom-left LED?          |        |
| 5. How many bytes for one full picture?          |        |

**Answers**

Cover! 1: 64 • 2: 8 (one byte) • 3:  
00011000 • 4: (7, 0) • 5: 8 bytes





## Objective check

# Can I really do it? Prove it.



**01** Point to LED (2, 5) on the matrix. How did you find it?



**02** Read out your emoji's 8 rows of bits.



**03** Show your animation. How many frames, and what waits did you use?



**04** Show your scrolling sign. Who would it help?



### Rate yourself

Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

**Teacher:** Emoji guessing game: show your emoji for 2 seconds. Can the class guess it?





Show what you know

# What did we learn?

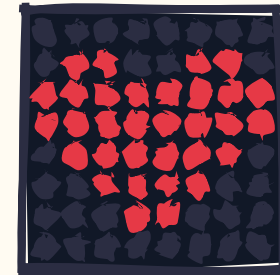
- ✓ Every LED has an address: (row, column), from 0 to 7
- ✓ Each row is 8 bits: one byte; a picture is 8 bytes
- ✓ Animation = frames shown one after another
- ✓ Scrolling shifts columns to show long messages
- ✓ The MAX7219 chip controls 64 LEDs with 3 wires



**Exit ticket:** write the 8 bits for a row where only the first and last LEDs are on.

## At home

Find an LED sign or scoreboard near you. Can you see the dots? Count how many rows it has.



## Next class:



Ultrasonic  
Theremin: Music  
in the Air!

